

Manzil JEE 2026

Chemistry (Practice Sheet)

Chemical Bonding

Time duration- 75 min.

Q1 Which of the following species are hypervalent ?

1. ClO_4^-
2. BF_3
3. SO_4^{2-}
4. CO_3^{2-}

- (A) 1, 2, 3 (B) 1, 3
(C) 3, 4 (D) 1, 2

Q2 The correct order of bond angles is

- (A) $\text{NO}_2^- > \text{NO}_2^+ > \text{NO}_2$
(B) $\text{NO}_2^+ > \text{NO}_2^- > \text{NO}_2$
(C) $\text{NO}_2 > \text{NO}_2^+ > \text{NO}_2^-$
(D) $\text{NO}_2^+ > \text{NO}_2 > \text{NO}_2^-$

Q3 Which pair among the following has identical bond order?

- (A) CN^- , CO^+
(B) O_2^- , O_2^+
(C) CN^- , CN^+
(D) NO^+ , CN^-

Q4 Which of the following are isoelectronic and isostructural? NO_3^- , CO_3^{2-} , ClO_3^- , SO_3

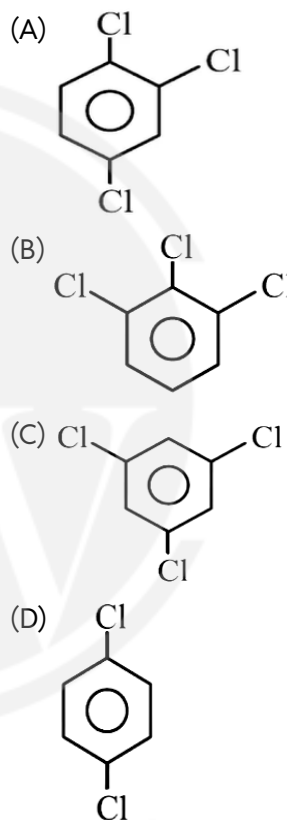
- (A) NO_3^- , CO_3^{2-}
(B) SO_3 , NO_3^-
(C) ClO_3^- , CO_3^{2-}
(D) ClO_3^- , SO_3

Q5 In PO_4^{3-} ion, the formal charge on each oxygen atom and P-O bond order respectively are

- (A) -0.75, 0.6 (B) -0.75, 1.0
(C) -0.75, 1.25 (D) -3, 1.25

Q6 Among LiCl , BeCl_2 , BCl_3 and CCl_4 , the covalent bond character follows the order

- (A) $\text{LiCl} < \text{BeCl}_2 > \text{BCl}_3 > \text{CCl}_4$

(B) $\text{LiCl} > \text{BeCl}_2 < \text{BCl}_3 < \text{CCl}_4$ (C) $\text{LiCl} < \text{BeCl}_2 < \text{BCl}_3 < \text{CCl}_4$ (D) $\text{LiCl} > \text{BeCl}_2 > \text{BCl}_3 > \text{CCl}_4$ **Q7** Which has maximum dipole moment?**Q8** The correct arrangement of the species in the decreasing order of the bond length between carbon and oxygen in them is

- (A) CO , CO_2 , HCO_2^- , CO_3^{2-}
(B) CO_2 , HCO_2^- , CO , CO_3^{2-}
(C) CO_3^{2-} , HCO_2^- , CO_2 , CO
(D) CO , CO_3^{2-} , CO_2 , HCO_2^-

Q9 The shape of XeF_4 , XeF_5^- and SnCl_2 are

- (A) octahedral, trigonal bipyramidal and bent
(B)



square pyramidal, pentagonal planar and linear

- (C) square planar, pentagonal planar and angular
(D) see-saw, T-shaped and linear

Q10 The solubility of KCl is relatively more in

- (A) C_6H_6 (Dielectric constant = 0)
(B) $(CH_3)_2CO$ (Dielectric constant = 2)
(C) CH_3OH (Dielectric constant = 32)
(D) CCl_4 (Dielectric constant = 0)

Q11 Which of the following has maximum % of s-character?

- (A) N_2H_2
(B) N_2H_4
(C) NH_3
(D) NH_2^-

Q12 A square planar complex is formed by hybridisation of which atomic orbital?

- (A) s, p_x, p_y, d_{yz}
(B) $s, p_x, p_y, d_{x^2-y^2}$
(C) s, p_x, p_y
(D) s, p_x, p_y, d_{xy}

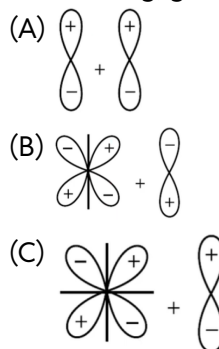
Q13 Consider the molecules CH_4 , NH_3 and H_2O . Which of the given statements is false?

- (A) The $H-O-H$ bond angle in H_2O is smaller than the $H-N-H$ bond angle in NH_3 .
(B) The $H-C-H$ bond angle in CH_4 is larger than the $H-N-H$ bond angle in NH_3 .
(C) The $H-C-H$ bond angle in CH_4 , the $H-N-H$ bond angle in NH_3 , and the $H-O-H$ bond angle in H_2O are all greater than 90° .
(D) The $H-O-H$ bond angle in H_2O is larger than the $H-C-H$ bond angle in CH_4 .

Q14 In which of the following shape is same but hybridization is different:

- (A) ICl_2^- , XeF_2
(B) SO_2 , NO_2^+
(C) SO_2 , NH_2^-
(D) CO_2 , SO_2

Q15 Which of the following orbital overlapping results antibonding, gerade M.O. having two nodal plane



(D) None of these

Q16 Which of the following species below have molecular geometry similar to $XeOF_4$?

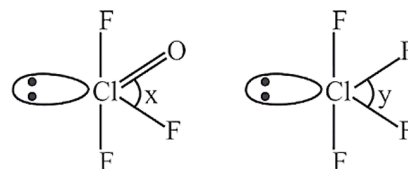
- (A) XeO_3 (B) PCl_5
(C) IOF_4^+ (D) XeF_5^+

Q17 Match the following items of column I with the appropriate items in column II.

Molecular species	Bond order and magnetic property
P. O_2^-	I. 2.5 paramagnetic
Q. O_2	II. 2.0 diamagnetic
R. O_2^+	III. 1.5 diamagnetic
S. O_2^{2-}	IV. 1.0 diamagnetic
	V. 2.0 paramagnetic
	VI. 1.5 paramagnetic

- (A) P-III, Q-V, R-IV, S-III
(B) P-III, Q-II, R-I, S-IV
(C) P-III, Q-V, R-IV, S-VI
(D) P-VI, Q-V, R-I, S-IV

Q18 Compare bond angle x and y in the following molecules



- (A) $x > y$ (B) $y > x$
(C) $x = y$ (D) None of these

Q19 Find the number of 90° angles in $XeF_5^\ominus$ molecules.



- (A) 1 (B) 2
(C) 0 (D) 3

Q20 Maximum seven atoms are present in a plane, Except in

- (A) H_2CSF_4
(B) $\text{P}(\text{CH}_3)_2(\text{CH}_3)_3$
(C) $\text{N}(\text{SiH}_3)_3$
(D) $\text{B}(\text{CH}_3)_3$

Q21 Choose the **incorrect** statement for CH_4 , BH_4^- , NH_4^+ :

- (A) They all have tetrahedral geometry.
(B) The $\text{H}-\text{X}-\text{H}$ (X : Central atom) angle follows the order $\text{CH}_4 > \text{BH}_4^- > \text{NH}_4^+$.
(C) They all are non-polar species.
(D) The central atom is sp^3 hybridised.

Q22 Which is **correct** option for overlapping strength:

- (A) $2s-2s < 2s-2p < 2p-2p$
(B) $1s-1s < 1s-2p < 2p-2p$
(C) $2s-2s < 1s-2p < 1s-1s$
(D) Both (A) and (C)

Q23 If the bond length and dipole moment of a diatomic molecule are $1.25 \text{ \AA}$ and 1.0 D respectively, what is the per cent ionic character of the bond?

- (A) 10.66 (B) 12.33
(C) 16.66 (D) 19.33

Q24 Which of the following molecule does not obey octet rule?

- (A) SF_6 (B) Cl_2
(C) NaCl (D) N_2

Q25 Which of the following orbital is having maximum number of nodal plane

- (A) σ_{p-p}
(B) π_{d-d} (2 lobes interaction)
(C) π_{p-p}^*
(D) π_{d-p}

Q26 For which value of x , the dipole moment of the molecule $\text{PCl}_x\text{F}_{5-x}$ is zero

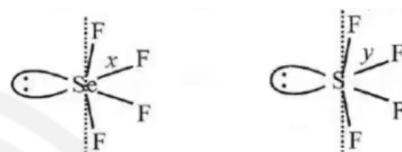
- (A) 4 (B) 2

- (C) 3 (D) 1

Q27 H_2 molecule is more stable than Li_2 molecule, because

- (A) in H_2 molecule, σ molecular orbitals are shielded by electrons
(B) in H_2 bond order is one
(C) in Li_2 molecule, σ molecular orbitals are shielded by electrons
(D) in Li molecule, outer $\sigma 2s$ molecular orbitals are shielded by the inner electrons

Q28 Compare bond lengths for the following molecules:



- (A) $x > y$ (B) $y > x$
(C) $x = y$ (D) None of these

Q29 Among KO_2 , AlO_2^- , BaO_2 & NO_2^+ unpaired electron is present in:

- (A) NO_2^+ and BaO_2
(B) KO_2 and AlO_2^-
(C) KO_2 only
(D) BaO_2 only

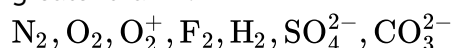
Q30 Which of the following statement is incorrect :-

- (A) Formal charge and bond order of NO_3^- is $-\frac{2}{3}$ and $\frac{4}{3}$
(B) XeF_4 is formed in first excited state
(C) Melting point of NaCl more than LiCl
(D) Solubility of LiF less than LiI

Q31 The hybridization state of central iodine atom in IF_5 , I_3^- and I_3^+ are respectively:

- (A) sp^3d^2 , sp^3d and sp^3
(B) sp^3d , sp^3d and sp^3
(C) sp^3d^2 , sp^3d^2 and sp^3
(D) sp^3d , sp^3d and sp^2

Q32 How many of the following have bond order greater than 1?



- Q33** How many of the following species have coordinate bond?
 CO_3^{2-} , NO_3^- , NO_2^- , BF_4^- , NH_4^+
- Q34** How many of following substance is having higher lattice energy than NaBr?
 CaCl_2 , NaI, CsBr, LiF, MgO, Al_2O_3 , TiO_2
- Q35** The formal charge on the carbon atom in the carbonate ion is
- Q36** How many of the following molecules have bond angle greater than 100° ?
 NH_3 , H_2O , H_2S , PH_3 , AsH_3 , OF_2 , OCl_2 , BF_3
- Q37** Consider the following molecules:
 SO_3 , SOCl_2 , XeO_2F_2 , HClO_4 , . Find
 H_2SO_3 , H_3PO_4
the value of $a + b + c$, where
 a = Total no. of $p\pi - d\pi$ bonds present in all the molecules
 b = No. of molecules with atleast one lone pair of electrons on central atom
 c = No. of non-polar molecules
- Q38** For the following molecules:
 PCl_5 , BrF_3 , ICl_2^- , XeF_5^- , NO_3^- ,
 XeO_2F_2 , PCl_4^+ , CH_3^+
Calculate the value of $\frac{a+b}{c}$
 a = Number of species having sp^3 d-hybridisation
 b = Number of species which are planar
 c = Number of species which are non-planar
- Q39** How many of the following have sp^3 hybridization?
 PF_3 , I_3^+ ; SiCl_4 ; SO_4^{2-} ; ClO_4^- ; NO_2^- ; SF_6 , IF_7
- Q40** In how many of the following molecules at least 4 atoms lie in same plane?
(I) XeOF_4 (II) BrF_3
(III) SF_4 (IV) XeO_3
(V) XeF_4 (VI) SOCl_2
(VII) PF_3 (VIII) NH_4^+

Q41

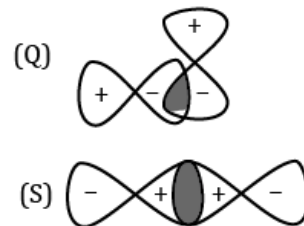
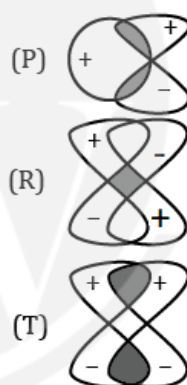
Find the number of molecule(s) in which $2p\pi - 2p\pi$ back bonding is present.
 $\text{N}(\text{SiH}_3)_3$, BF_3 , OCl_2

- Q42** Among the given triatomic molecules/ions:
 BeCl_2 , N_3^- , N_2O , NO_2^+ , O_3 , SCl_2 , ICl_2^- , I_3^-
and XeF_2 , the total number of linear molecule(s)/ion(s) where the hybridization of the central atom does not have contribution from the d-orbital(s) is _____.
[Atomic number : S = 16, Cl = 17, I = 53 and Xe = 54]

- Q43** Find total number of polar molecules.

- (a) PF_3Cl_2 (b) SF_4
(c) PCl_5 (d) PCl_3F_2
(e) SF_6 (f) XeF_2

- Q44** How many of the following atomic orbital overlappings are not allowed?



- Q45** Number of molecules among the following having non-zero dipole moment is
 XeO_3 , SO_3 , SF_4 , SF_6 , H_2S , CS_2 , SO_2 , H_2O
- Q46** The number of pyramidal molecules from the following is:
 CH_3^+ , NH_3 , O_3 , SO_2 , CH_4 , XeO_3 .
- Q47** Find the number of species in which bond angle(s) at central atom is less than idealised bond angle as per hybridisation.



NOF,
 XeF_4 , XeF_3^+ , SbCl_5^{2-} , IF_2^+ , XeF_5^- , O_3 ,
 $\text{O}(\text{CH}_3)_2$

Q48 Consider the following compounds:

- (i) IF_5
- (ii) ICl_4^-
- (iii) XeO_2F_2
- (iv) NH_2^-
- (v) BCl_3
- (vi) BeCl_2

Then, calculate value of $x + y - z$, here x , y and z are total number of compounds in given compounds in which central atom used their all three p-orbitals, only two p-orbitals and only one p-orbital in hybridisation respectively.

Q49 How many of the following are true statements?

- (a) CO has much less dipole moment (0.11D) than expected.
- (b) SiCl_4 has abnormally low boiling point than CCl_4
- (c) Hybridisation in $(\text{CH}_5)^+$ is sp^2 but here carbon is pentavalent
- (d) B_2 and C_2 are examples of species having no sigma but only "pi" bonds
- (e) For $\text{CF}_3 - \text{CF}_2 - \text{CF}_3$, two C – F bonds attached to the middle carbon atom are longer as compared to the other C – F bonds at the terminal carbons.

Q50 The number of correct statements among the following is

- (I) $(\text{CH}_3)_3\text{N}$ is pyramidal whereas $(\text{SiH}_3)_3\text{N}$ is planar
- (II) PCl_5 is super octet molecule
- (III) $\text{p}\pi - \text{d}\pi$ bonds are present in SO_2
- (IV) Hybridisation of central atom is same in both SF_4 and ICl_3
- (V) I_3^- has linear structure
- (VI) SeF_4 and CH_4 have same shape
- (VII) Central atom has same number of lone pair(s) in both SeF_4 and XeF_6



Answer Key

Q1 (B)
Q2 (D)
Q3 (D)
Q4 (A)
Q5 (C)
Q6 (C)
Q7 (B)
Q8 (C)
Q9 (C)
Q10 (C)
Q11 (A)
Q12 (B)
Q13 (D)
Q14 (C)
Q15 (D)
Q16 (D)
Q17 (D)
Q18 (B)
Q19 (C)
Q20 (A)
Q21 (B)
Q22 (D)
Q23 (C)
Q24 (D)
Q25 (B)

Q26 (C)
Q27 (D)
Q28 (A)
Q29 (C)
Q30 (B)
Q31 (A)
Q32 5
Q33 3
Q34 5
Q35 0
Q36 5
Q37 14
Q38 3
Q39 5
Q40 3
Q41 1
Q42 4
Q43 2
Q44 3
Q45 5
Q46 3
Q47 5
Q48 4
Q49 5
Q50 6

